

CHARLES BENELLO

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EDUCATION

The University of Chicago

Chicago, IL

Master of Science in Financial Mathematics

Expected Mar 2027

- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: ML in SWE, ML in Finance, Agentic AI, Distributed Systems, Reinforcement Learning

The University of Chicago | GPA: 3.75/4.0

Chicago, IL

Bachelor of Science in Mathematics and Computer Science, Minor in Korean

Jun 2025

- Coursework: Database Systems, Operating Systems, Algorithms, Concurrency Control, Computer Architecture, Networks

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Go, TypeScript, JavaScript, SQL, CUDA

Frameworks & Tools: PyTorch, React, Next.js, Flask, FastAPI, React Native, Docker, Git, Linux, PostgreSQL, DuckDB, Databricks, XGBoost

Concepts: Distributed Systems, Database Internals, Algorithms, Data Structures, Concurrency, GPU Programming, ML, System Design

EXPERIENCE

Software Engineering Intern

Feb 2026 – Present

Alter Domus

Chicago, IL

- Lifted document extraction accuracy from 50% to 92–97% by architecting a multi-stage pipeline with schema-enforced structured output and cross-reference verification for 100+ page financial documents.
- Reduced hallucination rates across the platform's AI systems by building an LLM-as-a-Judge evaluation framework with automated benchmarks for faithfulness, numerical accuracy, and completeness.
- Cut LLM inference costs and improved response speed by optimizing context handling, adding personalization to client-facing AI, and removing context length barriers for large document ingestion.

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored VLDB 2026 paper by designing a parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort.
- Sole-authored HPTS 2026 paper by building a lightweight prediction model for sort configuration on cloud/disaggregated hardware, transferring calibration across machines within 3–5% of optimal.
- Improved random access speed by 200% and reduced memory footprint by 40% by engineering Small String Optimization across storage structures, inlining short keys directly into nodes to eliminate heap allocations and pointer chases on the hot path.

Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Built a self-supervised transaction foundation model (TabBERT-style hierarchical 2-level transformer, ~5M params) on 14.5M unlabeled transactions; achieved AUC 0.974 / F1 0.822 on downstream fraud detection at 0.12% positive rate.
- Ran 87+ ablations on UChicago's Midway3 SLURM cluster; implemented per-field masking and SPRM from PANTHER (NeurIPS 2025) and PRAGMA (Revolut 2026), with per-field masking alone yielding +1.9% AUC and +16.8% F1.

Quantitative Researcher

Oct – Dec 2025

Bank of America (UChicago Project Lab)

Chicago, IL

- Built an AI-driven platform to rebalance high-quality liquid asset portfolios, generating trade recommendations that minimize risk exposure over 6-month projections under capital, liquidity, and net interest income constraints.
- Shipped full-stack interfaces (Next.js, Flask, React Native, FastAPI) and engineered a multi-agent LLM debate system where competing agents generate macro stress scenarios and produce probability-weighted reallocation recommendations.

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab)

Lausanne, Switzerland

- Cut out-of-memory query times by 50–200% by engineering a macroblock-based GPU decompression strategy in CUDA/C++ for large analytical workloads.
- Drove a new cost-based compression research direction by benchmarking GPU compression techniques with resource contention models and building performance models enabling adaptive decompression on TPC-DS workloads.

PROJECTS

Transit Optimizer | Python | [cmbenello.github.io/covtsp](https://github.com/cmbenello/covtsp)

2026

- Beat the Guinness World Record by 60 min (16h45m vs. 17h45m) by building a fully automated Covering TSP pipeline: given a city name, fetches GTFS feeds and Google Maps walking data, builds a time-expanded graph (160K nodes, 1.26M edges), and runs k-NN with urgency scoring, hard-station auto-detection, and epsilon-greedy search. Generalized to NYC (475 stations) and Berlin.

MiniMetro RL | Python, PyTorch | [cmbenello.github.io/MiniMetro](https://github.com/cmbenello/MiniMetro)

2025

- Built a custom Gymnasium environment simulating the transit-building game Mini Metro, with a full game engine (stations, lines, passengers, routing) and trained a PPO agent to learn network construction strategies.

Relational Database Engine | Rust | github.com/cmbenello/CrustyDB

2023

- Implemented a multi-threaded relational database from scratch: B+ tree indexing, concurrent buffer pool management with page-level locking, heap storage, and a query execution engine supporting SQL operations.

PUBLICATIONS

[1] R. Otaki*, C. Benello*, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (*Equal contribution)

[2] C. Benello. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *HPTS 2026*.

[3] R. Otaki, J.H. Chang, C. Benello, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025*.

ADDITIONAL

Languages: Korean, Italian (Intermediate); French (Basic) | **Interests:** Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting